INFO 4220 – Networks II: Market Design Spring 2022 – Problem Set E

Problem 1 (16 points)

There are six students (Antonio, Bruno, Camilo, Dolores, Julieta, and Mirabel) that need to enroll in up to two courses chosen out of 5 options: $c_1, c_2, \dots, c_5$. Courses c_1, c_2, c_4 have a capacity of 2, while c_3, c_5 have a capacity of 3. There is a time conflict between course c_2 and c_3 and thus students cannot enroll in those two courses together. Each student is given a budget of 1000 tokens to bid for courses.

The preferences of students over course are:

$P_{Antonio}$	P_{Bruno}	P_{Camilo}	$P_{Dolores}$	$P_{Julieta}$	$P_{Mirabel}$
c_1	c_2	c_3	c_4	c_3	c_5
c_2	c_3	c_2	c_2	c_4	c_1
c_3	c_4	c_5	c_5	c_5	c_2
c_4	c_5	c_4	c_3	c_1	c_3
c_5	c_1	c_1	c_1	c_2	c_4

Assume that students submit the following bids for courses:

	Antonio	*Bruno*	*Camilo*	*Dolores*	*Julieta*	*Mirabel*
c_1	202	209	211	204	201	245
c_2	260	207	180	206	200	150
c_3	257	121	259	144	350	244
c_4	243	205	100	238	226	136
c_5	38	258	250	208	23	225

a) Find the course allocation if the school uses an auction (course bidding) mechanism (6 pts)

The first thing we need to do is order the bids from highest to lowest

Bid	*Student*	*Course Selected*
350	*Julieta*	c_3
260	*Antonio*	c_2
259	*Camilo*	c_3
258	*Bruno*	c_5
257	*Antonio*	C_3 *(Cannot register because of time conflict)*
250	*Camilo*	C_5
245	*Mirabel*	C_1
244	*Mirabel*	C_3
243	*Antonio*	C_4
238	*Dolores*	C_4

226	*Julieta*	C_4 *(But cannot register because course is full)*
225	*Mirabel*	*Has already filled her schedule*
211	*Camilo*	*Has already filled his schedule*
209	*Bruno*	C_1
208	*Dolores*	C_5
207	*Bruno*	*Has already filled his schedule*
206	*Dolores*	C_2
205	*Bruno*	*Has already filled his schedule*
204	*Dolores*	*Has already filled her schedule*
202	*Antonio*	*Has already filled his schedule*
201	*Julieta*	C_1 *(Cannot register because course is full)*
200	*Julieta*	C_2 *(Cannot register because of time conflict)*
180	*Camilo*	*Already filled his schedule*
150	*Mirabel*	*Already filled her schedule*
144	*Dolores*	*Already filled her schedule*
136	*Mirabel*	*Already filled her schedule*
121	*Bruno*	*Already filled his schedule*
100	*Camilo*	*Already filled his schedule*
38	*Antonio*	*Already filled his schedule*
23	*Julieta*	C_5 *(Cannot register because course is full)*

At this point, everyone has 2 courses:

$Antonio$	$Bruno$	$Camilo$	$Dolores$	$Julieta$	$Mirabel$
c_2, c_4	c_1, c_5	c_3, c_5	c_4, c_5	c_3	c_1, c_3

b) What is the price of each of the courses? Is there any student that didn't get a course that they liked better than one of the courses that they did get, even though they had bid an amount that is over the price of such course? (2 pts)

c_1	*209*
c_2	*0*
c_3	*244*
c_4	*238*
c_5	*208*

Mirabel liked c_5 the most. She had bid 225 for c_5 and the clearing price for c_5 ended up being 208 and yet she did not get the course. This happened because her bids for c_3, c_1 were processed before her bid for c_5.

c) Find the course allocation that would result if the school used the Gale-Shapley Pareto-Dominant Market Mechanism (6 pts)

Step 1:

Antonio proposes to c_1, c_2

Bruno proposes to c_2, c_4 *(note that Bruno likes* c_3 *better than* c_4*, but has a time conflict. Thus, proposes to* c_4*)*

Camilo proposes to c_3, c_5 *(note that Camilo likes* c_2 *better than* c_5*, but has a time conflict. Thus, proposes to* c_5*)*

Dolores proposes to c_2, c_4

Julieta proposes to c_3, c_4

Mirabel proposes to c_1, c_5

c_2 *received 3 proposals and has only 2 spots, it rejects the proposal with the lowest bid: Dolores*

c_4 *received 3 proposals and has only 2 spots, it rejects the proposal with the lowest bid: Bruno*

Step 2:
Dolores now proposes to c_5

Bruno now proposes to c_5

Now c_5 *has 4 offers and only 3 spots. It rejects the proposal with the lowest bid: Dolores*

Step 3:

Dolores proposes to c_3*. No course is over capacity and thus algorithm ends.*

$Antonio$	$Bruno$	$Camilo$	$Dolores$	$Julieta$	$Mirabel$
c_1, c_2	c_2, c_5	c_3, c_5	c_3, c_4	c_3, c_4	c_1, c_5

d) What is the price of each of the courses? Is there any student that didn't get a course that they liked better than one of the courses that they did get, even though they had bid an amount that is over the price of such course? (2 pts)

c_1	*202*
c_2	*207*
c_3	*144*
c_4	*226*
c_5	*225*

No student bid an amount that was above the price of a course that they didn't get and liked better to a course that they got.

Problem 2 (17 points)

Assume an auction with n bidders, each with valuation v_i that is drawn from a uniform distribution on $[50,200]$. For this question, do not use any result from class. Calculate everything explicitly and show all your steps.

a) Find the expected revenue if there are only 2 bidders and an ascending price auction is used (ignore ticking prices, and assume that a buyer prefers to buy the item rather than not if the price is equal to their valuation) (3 pts)

The expected value of the minimum between two random draws out of uniform distribution on [50, 200] is 100. What will happen is that the bidder with the lowest valuation will bid until the price reaches 100. At that point they drop and bidder 2 wins and pays 100. The expected revenue is thus 100

b) The seller is unhappy with the expected revenue in (a) and decides to use a reserve price r. To do so, the English auction will start at a price r. Calculate the probability that the item sells if the seller uses a reserve price r and the expected revenue (Still assume there are 2 bidders). Find the optimal reserve price. (6 pts)

There are 3 scenarios that could happen depending on the valuation drawn by the bidders:

Case 1: Both bidders draw values below the reserve price. In this case the item doesn't sell and the expected revenue is 0. This happens with probability:

P$(v_1 < r) * P(v_2 < r) = \left(\frac{r-50}{150}\right)^2$

Case 2: One of the bidders draw a valuation below r, and one of the bidders draw a valuation above r. In this case the item sells for r. This happens with probability:

P$(v_1 < r) * P(v_2 > r) + P(v_1 > r) * P(v_2 < r) = 2 * \frac{r-50}{150} * \frac{200-r}{150}$

Case 3: Both bidders draw valuations above r. In this case the item will sell for the valuation of the bidder with the lowest valuation. This is trickier to consider. The probability that this scenario happens is:

$P(v_1 > r) * P(v_2 > r) = \left(\frac{200-r}{150}\right)^2$

We now need to consider what will be the expected value of the lowest between two draws given that we know both are at least r. You can think of this as drawing two values from a uniform distribution between (r,200). Thus, the expected value of the minimum between the two random draws is:
$r + \frac{1}{3} * (200 - r) = \frac{2}{3}r + 66.67$

The probability that the item sells is: $2 * \frac{r-50}{150} * \frac{200-r}{150} + \left(\frac{200-r}{150}\right)^2 = \frac{20{,}000+100r-r^2}{150^2}$

*The expected revenue is: 2** $(\frac{r-50}{150} * \frac{200-r}{150}) * r + \left(\frac{200-r}{150}\right)^2 * (\frac{2}{3}r + \frac{200}{3})$

To find the optimal reserve price we can maximize the expected revenue. We take the derivative of expected revenue and equal it to zero: $-4r^2 + 600r - 20{,}000 = 0 \Leftrightarrow 4 * (r - 50)(r - 100) = 0$. *It is easy to see the optimal reserve price is r = 100*

Now assume that there are 2 bidders. Bidder's 1 valuation is drawn from a uniform distribution on [0, 100], while Bidder's 2 valuation is drawn from a uniform distribution on [0, 200]. Assuming that we use an English auction (ignore ticking prices and assume that a buyer prefers to buy the item rather than not if the price is equal to their valuation).

c) Calculate the probability that each bidder wins and the price the item would sell for in each case. Then compute the expected revenue of this auction (4 pts)

We can think in two scenarios that are equally like (prob. ½ each):

Case 1: Bidder 2 draws a valuation that is above 100. In this case bidder 2 wins for sure and is going to pay the valuation of bidder 1. The expected value of the valuation of bidder 1 in this case is the expected value of 1 random draw from a uniform distribution on (0, 100), which is 50. Thus, in this case Bidder 2 wins and pays 50.

Case 2: Bidder 2 draws a valuation that is between (0,100). This also has a probability of happening of ½. We can think of this scenario as an auction between 2 bidders with valuations drawn from a uniform distribution on (0, 100). In this case, each bidder has a probability ½ of winning, and they will pay the expected value of the lowest between 2 random draws from a uniform distribution on (0, 100), which is 33.33.

Thus:
Bidder 2 wins in ¾ of the cases, and bidder 1 wins in ¼ of the cases.
*The expected revenue of the auction is 0.5*50+0.5*0.5*100/3+0.5*0.5*100/3=41.67*

d) The auctioneer decides to offer bidder 1 a discount. If bidder 1 bids b, then they only have to pay b/2. What is the optimal bidding strategy for each bidder in this auction? Find the probability that each bidder wins the auction, the price they pay, and the expected revenue in the auction. Is offering this discount to bidder 1 a good strategy? Why or why not? (4 pts)

The optimal bidding strategy for bidder 1 is to bid $2 * v_1$, *and for bidder 2 is to bid* v_2. *Thus, the auction becomes alike an auction with 2 bidders each drawing their valuations from a uniform distribution on (0, 200). Each bidder will win with probability 0.5 with a bid of 66.67. However, when bidder 1 wins, they will only pay half their bid. Thus, the expected revenue in this auction is 0.5 * 33.33 + 0.5 * 66.67 = 50. It is a good strategy for the auctioneer because the expected revenue is higher. This happens because by offering a discount to bidder 1, it makes the auction more competitive forcing bidder 2 to bid higher.*

Problem 3 (17 points)

a) Consider a VCG auction with 5 bidders and 2 objects. The bids submitted are as follows:

	A	B	A,B
Bidder 1	28	0	0
Bidder 2	0	20	0
Bidder 3	0	0	32
Bidder 4	12	0	0
Bidder 5	0	11	0

Find who will get which object and at what price (5 pts)

The assignment that maximizes social value is to give object A to bidder 1 and object B to bidder 2. To determine how much each of them must pay, we need to compute the externality they impose on other bidders.

If bidder 1 was not participating in the auction, then bidder 4 would get object A and obtain a value of 12. When bidder 1 participates, bidder 4 doesn't get any object and obtains a value of 0. Thus, the price bidder 1 must pay is 12. (note that an equivalent reasoning can be made saying that when bidder 1 is absent both objects are assigned to bidder 3).

I bidder 2 was not participating, then bidder 5 would get object B and obtain a value of 11. When bidder 2 participates bidder 5 doesn't get any object and obtains a value of 0. Thus, bidder 2 must pay 11 for object B.

b) Consider a VCG auction with 3 bidders and 2 objects. The bids submitted are as follows:

	A	B	A,B
Bidder 1	0	5	5
Bidder 2	5	0	5
Bidder 3	0	0	5

Find who will get which object and at what price. In this auction, would any bidder have incentives to try to convince the auctioneer to make a deal and deviate from the outcome determined by the VCG auction (6 pts)

In this case we should assign the object B to bidder 1, and object A to bidder 2.

To determine how much should bidder 1 pay, let's see what happens when bidder 1 doesn't participate in the auction. If bidder 1 doesn't participate, we can assign both objects to bidder 3 or to bidder 2, or just object A to bidder 2. In all cases bidders 2 and 3 (collectively) obtain a value of 5. When bidder 1 participates, bidder 2 gets object A and obtains a value of 5. This means the price bidder 1 must pay is 0. Same reasoning applies to bidder 2 that also pays 0.

Note that the auctioneer is probably not very happy about this outcome (as they don't get any money), and so is bidder 3 (that doesn't an object). If Bidder 3 and the auctioneer make a deal in which dealer 3 gets both objects and pays something (between 0-5), the auctioneer and bidder 3 would be better off.

c) Consider a VCG auction with two bidders that are bidding for two identical units of one good. Bidder 1 submits a bid of $5 for one unit (they want only one unit), and bidder 2 submits a bid of $5 for one unit and $9 for two units. Find how many units each bidder gets and at what price. Is the allocation envy-free? (6 pts)

In this case bidder 1 gets one unit, and bidder 2 gets one unit. The prices they pay is bidder 1 pays 4, while bidder 2 pays 0. This is not envy-free, because bidder 1 would like to get the object that bidder 2 got (which is exactly the same object they got) at the price that bidder 2 paid (which is 0 instead of 4).